

ADAM v32: Master Hardware Blueprint

This document outlines the final, crash-free architecture for ADAM. It divides the robot into two physical zones (Body and Head) to eliminate DMA bottlenecks and drastically reduce the number of wires passing through the mechanical neck joint.



1. Final Component List

IN THE BODY (The Base)

1. **Raspberry Pi Zero 2W:** The main brain. Runs Gemini, Audio, and OpenCV.
2. **INMP441 Microphones (x2):** Stereo hearing via I2S.
3. **MAX98357A Amplifier + 3W Speaker:** Voice output via I2S.
4. **RCWL-0516 Microwave Radar:** Hidden inside the plastic body. Wakes ADAM up when someone walks into the room. (*Replaces the LiDAR*).
5. **MG90S Servo (Pan):** Moves the neck left and right.
6. **5V Buck Converter:** Steps down battery voltage to power everything safely.

IN THE HEAD (The Moving Part)

7. **ESP32-CAM:** Handles Video streaming, reading touch sensors, and moving the head up and down.
8. **Raspberry Pi Pico (or Pico W):** Dedicated "Graphics Card" solely for rendering ADAM's face at 30 FPS.
9. **ILI9341 TFT Display (2.42"):** ADAM's face.
10. **PCF8574T I2C Expander:** Groups the 4 touch sensors into just 2 pins.
11. **TTP223 Touch Sensors (x4):** For petting/touch interactions.
12. **MG90S Servo (Tilt):** Moves the head up and down.



2. The Two Communication Flows

You must choose one of these two methods for how the devices talk to each other.

FLOW 1: The "Wi-Fi Relay" Method (🏆 Highly Recommended)

In this flow, the Pi uses Wi-Fi to request video and tell the ESP32-CAM to change the emotion. The ESP32-CAM then uses a 1-inch wire inside the head to pass that emotion command to the Pico.

- **Video:** ESP32-CAM ---(Wi-Fi)---> Pi Zero
- **Emotions:** Pi Zero ---(Wi-Fi)---> ESP32-CAM ---(TX wire)---> Pico
- **Wires passing through the Neck: Just 2 wires!** (5V and GND). Everything else is wireless or wired locally inside the head.

FLOW 2: The "Full Hardwired" Method (Zero Wi-Fi)

In this flow, Wi-Fi is completely disabled for an offline build. The Pi Zero uses a high-speed Serial (UART) connection to receive continuous JPEG video frames from the ESP32-CAM and to send emotion commands back to it. The ESP32-CAM then acts as a relay, passing the emotion commands via a direct local wire to the Pico.

- **Video & Commands:** ESP32-CAM <---(High-Speed UART)---> Pi Zero
- **Emotions Relay:** ESP32-CAM --- (Local Wire) ---> Pico
- **Wires passing through the Neck: 4 wires** (5V, GND, Pi TX, and Pi RX).

3. Master Pinout Connections

A. Raspberry Pi Zero 2W (Body)

Pi Physical Pin	GPIO	Connects To
Pin 12	GPIO 18	Mic SCK AND Amp BCLK (Shared Clock)
Pin 35	GPIO 19	Mic WS AND Amp LRC (Shared Clock)
Pin 38	GPIO 20	Mic SD (Data IN)
Pin 40	GPIO 21	Amp DIN (Data OUT)
Pin 32	GPIO 12	Pan Servo (Orange/Yellow Signal Wire)
Pin 36	GPIO 16	RCWL-0516 Radar OUT Pin
Pin 8	GPIO 14 (TX)	<i>Only for Flow 2:</i> Connects to ESP32-CAM RX (GPIO 3)
Pin 10	GPIO 15 (RX)	<i>Only for Flow 2:</i> Connects to ESP32-CAM TX (GPIO 1)

B. ESP32-CAM (Head)

ESP32-CAM Pin	Connects To
GPIO 14	PCF8574T I2C Expander SDA
GPIO 15	PCF8574T I2C Expander SCL
GPIO 13	Tilt Servo (Orange/Yellow Signal Wire)
GPIO 1 (U0TXD)	<i>Only for Flow 2:</i> Connects to Pi Zero RX (Pin 10)
GPIO 3 (U0RXD)	<i>Only for Flow 2:</i> Connects to Pi Zero TX (Pin 8)
GPIO 4 (Software TX)	Connects to Pico GP1 (RX) to relay emotion commands locally in the head

C. PCF8574T I2C Expander (Head)

Expander Pin	Connects To
SDA & SCL	ESP32-CAM GPIO 14 & 15
P0	Touch Sensor 1 Signal
P1	Touch Sensor 2 Signal
P2	Touch Sensor 3 Signal
P3	Touch Sensor 4 Signal

D. Raspberry Pi Pico (Head)

Note: The Pico is powered by the 5V rail into its VSYS or VBUS pin, and powers the TFT using its 3V3 OUT pin.

Pico Pin	Connects To
GP19 (Pin 25)	TFT MOSI
GP18 (Pin 24)	TFT SCLK / SCK
GP17 (Pin 22)	TFT CS
GP16 (Pin 21)	TFT DC
GP20 (Pin 26)	TFT RST
GP1 (Pin 2)	UART RX: Receives commands from ESP32-CAM GPIO 4